1. Introduction

This guide explains the complete installation procedure for deploying the AI-Powered SOC Assistant integrated with Wazuh SIEM.

The deployment consists of:

- Docker-based Wazuh Server
- Windows Endpoint Agent
- Python AI Monitor
- AI Analyzer

The guide follows the exact implementation performed during the project.

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2. System Requirements

Host Machine

Component	Requirement
Operating System	Kali Linux
RAM	Minimum 8 GB
CPU	Intel i5 / Ryzen 5 or above
Storage	Minimum 50 GB

Endpoint Machine

Component	Requirement
Operating System	Windows 11
RAM	4 GB or above

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3. Software Requirements

- Docker Desktop / Docker Engine
- Docker Compose
- Git

- Python 3.13
- Wazuh 4.12
- Windows Wazuh Agent
- Visual Studio Code (optional)

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4. Project Architecture

```
Windows Endpoint
   │
   ▼
Wazuh Agent
   │
   ▼
Dockerized Wazuh Manager
   │
   ▼
alerts.json
   │
   ▼
Python AI Monitor
   │
   ▼
AI Analyzer
   │
   ▼
SOC Analyst
```

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5. Docker Installation

Step 1

Update packages.

sudo apt update

Step 2

Install Docker.

sudo apt install docker.io

Step 3

Enable Docker.

sudo systemctl enable docker

Step 4

Start Docker.

sudo systemctl start docker

Step 5

Verify Docker installation.

docker ps

Expected Result:

Docker runs successfully without errors.

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6. Wazuh Deployment

Step 1

Clone Wazuh Docker repository.

git clone https://github.com/wazuh/wazuh-docker.git

Step 2

Navigate to the deployment directory.

cd wazuh-docker/single-node

Step 3

Generate SSL certificates.

docker compose -f generate-indexer-certs.yml run --rm generator

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Step 4

Deploy Wazuh.

docker compose up -d

Step 5

Verify containers.

docker ps

Expected containers:

- Wazuh Manager
- Wazuh Dashboard
- Wazuh Indexer

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7. Accessing Wazuh Dashboard

Open the browser.

https://<Manager-IP>

Example:

https://192.168.xx.xxx

Login using administrator credentials.

Verify:

- Dashboard opens successfully.
- Manager is healthy.

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8. Windows Agent Installation

Download the Wazuh Agent MSI installer.

Install the agent on the Windows system.

During installation specify:

- Wazuh Manager IP
- Agent Name

Example:

Manager IP:
192.168.xx.xxx

Agent Name:
Balaganesh

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9. Agent Registration

Open Command Prompt as Administrator.

Navigate to:

cd "C:\Program Files (x86)\ossec-agent"

Authenticate the agent.

agent-auth.exe -m <Manager-IP>

Expected Output:

Valid key received

Start the service.

NET START WazuhSvc

Expected Output:

The Wazuh service was started successfully.

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10. Agent Verification

Login to Wazuh Dashboard.

Navigate to:

Agents

Verify:

- Agent is Active
- Agent Status = Connected

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11. Python Environment Setup

Create project directory.

mkdir AI_Wazuh_Lab

Navigate to project.

cd AI_Wazuh_Lab

Create virtual environment.

python3 -m venv venv

Activate environment.

source venv/bin/activate

Install dependencies.

pip install openai python-dotenv

Verify packages.

pip list

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12. AI Module Setup

Create the following files.

ai_monitor.py

ai_analyzer.py

config.py

.env

Store the OpenAI API key (if using OpenAI) or configure the local AI analyzer according to your implementation.

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13. Running the AI Monitor

Execute:

python ai_monitor.py

Expected Output:

===== Latest Wazuh Alert =====

Agent:
Balaganesh

Rule ID:
60106

Threat Category:
Authentication

Risk Level:
Low

MITRE:
T1078

Recommendations

Review activity

Continue monitoring

Confidence:
92%

The monitor should continue running and automatically process new alerts as they are generated.

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14. Testing

Generate Windows events:

- Successful Login

- Failed Login
- Logoff
- Account Lock
- PowerShell Execution (optional)

Verify:

- Alert appears in Wazuh Dashboard.
- AI monitor detects the alert.
- AI analysis is displayed.

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15. Troubleshooting

Docker Containers Not Running

Check:

docker ps

Restart:

docker compose restart

Dashboard Not Opening

Verify:

- Docker containers are running.
- Correct Manager IP is used.
- Browser certificate warning is accepted (if using self-signed certificates).

Windows Agent Not Connected

Verify:

NET START WazuhSvc

Re-authenticate:

agent-auth.exe -m <Manager-IP>

AI Monitor Shows JSON Error

Cause:

Incomplete alert written to alerts.json.

Solution:

Ignore partial entries and continue monitoring.

OpenAI API Error

Possible causes:

- Invalid API key (401)
- Insufficient quota (429)

Solution:

Verify API configuration or use the local AI analyzer until API access is available.